7 November 2026

Dear Editors,

Please consider our manuscript, “Tidal potential measured with distant optical clocks”, for publication as an Article in *Nature*.

Optical clocks measure gravitational potential differences through their relative ticking rates. An important next step for clock networks is to identify individual contributions to a changing potential, so that a measured background can be removed when searching for weaker signals. Our experiment addresses this problem by resolving neighbouring lunar and solar-frequency semidiurnal contributions in a distant clock comparison.

We compare strontium and ytterbium clocks in coastal Shanghai and inland Wuhan, separated by 692 km and connected by a 1,350-km fibre link. The record contains 545.9 hours of observations distributed over 353 days. The full tidal response is 0.96 ± 0.11 relative to the combined solid-Earth and ocean-loading prediction. The equivalent-height amplitude spectrum has peaks of 4.87 and 2.87 cm near the principal lunar and solar semidiurnal frequencies. Conditional spectra, reciprocal subtraction and correlation analysis support the separate identification of these contributions. Observations across changing lunar–solar phase relationships also allow us to quantify how solar-day-synchronous variations affect the lunar estimate.

The advance is the experimental identification of distinct time-dependent potential contributions using a clock network. Reciprocal subtraction and sampling-response calculations trace selected sidelobes to the lunar sampling response. These checks connect frequency metrology with the practical problem of separating geophysical backgrounds. We believe this result will interest readers working on precision measurement, gravitation and Earth observation, as clock networks develop towards sustained measurements of weaker potential changes.

The summer record includes 274 hours used in an earlier absolute-frequency-ratio study. The present manuscript identifies this shared record and reports the analysis of time-dependent tidal potential and constituent separation. An independent study has also reported tidal potential variations in a European optical-clock network: K. Stahl et al., *Optical clock network tracks tidal variations of the gravity potential*, arXiv:2610.39635v1 (2026).

Yours sincerely,

The authors